

Macro variables #8550 - #8564 let a G-code program obtain information about individual tools. When you specify an individual tool ID number with macro #8550, the control returns the individual tool information in macro variables #8551 - #8564. You can also specify an ATM group number with macro #8550. In this case, the control returns the individual tool information for the current tool in the specified ATM tool group in macro variables #8551 - #8564. Refer to the description for variables #8550 - #8564 in the Macros chapter. The values in these macros provide data that is also accessible from macros starting at 1601, 1801, 2001, 2201, 2401, 2601, 3201, and 3401 and for macros starting at 5401, 5501, 5601, 5701, 5801, and 5901. These provide access for tool data for tools 1-99. Macros #8551 - #8564 give access to the same data, but for tools 1-99 for all data items.

Save Advanced Tool Management Tables

You can save to USB the variables associated with Advanced Tool Management (ATM).

To save the ATM information:

1. Select the USB device in the Device Manager (**[LIST PROGRAM]**).
2. Type a filename on the input line.
3. Press **[F4]**.
4. Highlight **SAVE ATM** in the pop-up menu.
5. Press **[ENTER]**.

Restore Advanced Tool Management Tables

You can restore from USB the variables associated with Advanced Tool Management (ATM).

To restore the ATM information:

1. Select the USB device in the Device Manager (**[LIST PROGRAM]**).
2. Press **[F4]**.
3. Highlight **LOAD ATM** in the pop-up menu.
4. Press **[EMERGENCY STOP]**.
5. Press **[ENTER]**.

4.10 Tool Turret Operations

To operate the tool turret, refer to the following sections: Air Pressure, Eccentric Locating Cam Buttons, Protective Cap, and Tool Load or Tool Change.

4.10.1 Air Pressure

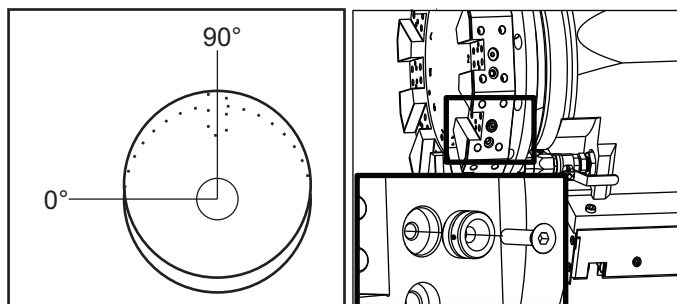
Low air pressure or insufficient air volume reduces the pressure applied to the turret clamp/unclamp piston. This can slow down the turret index time, or the turret may not unclamp.

4.10.2 Eccentric Locating Cam Buttons

Bolt-on turrets have eccentric locating cam buttons that let you finely align your ID tool holders to the spindle centerline.

Mount the tool holder to the turret and align the tool holder to the spindle in the X-axis. Measure the alignment in the Y-axis. If necessary remove the tool holder and use a narrow tool in the cam button hole, to rotate the eccentric to correct misalignment.

T4.3: This table gives the result for specific positions of the cam button.

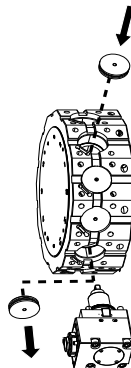


Rotation (degrees)	Result
0	no change
15	0.0018" (0.046 mm)
30	0.0035" (0.089 mm)
45	0.0050" (0.127 mm)
60	0.0060" (0.152 mm)
75	0.0067" (0.170 mm)
90	0.0070" (0.178 mm)

4.10.3 Protective Cap

IMPORTANT: Insert protective caps into empty turret pockets to protect them from accumulating debris.

F4.18: Turret Protective Caps in Empty Pockets



4.10.4 Tool Load or Tool Change

To load or change tools:



NOTE:

Y-Axis lathes return the turret to the zero position (spindle centerline) after a tool change.

1. Enter **MDI** mode.
2. Optional: Type the tool number that you want to change to in the format **Tnn**.
3. Press **[TURRET FWD]** or **[TURRET REV]**.

If you specified a tool number, the turret indexes to that turret position. Otherwise, the turret indexes to the next or the previous tool.

4.10.5 Hybrid Turret, VDI, and BOT Centerline Offset

To set the X offset to centerline for tools:

1. Press **[HANDLE JOG]** and enter the **Tool Geometry** offset page.
2. Select the **x offset** column and press **[F2]**.

For BOT (Bolt-On) turrets: Pressing **[F2]** sets an X-axis I.D. Tool Offset on center for a 1" (25 mm) I.D. BOT tool. Adjust offset manually for other size tooling or after-market toolholders.

For VDI (Verein Deutscher Ingenieure) turrets: Pressing **[F2]** sets an X-axis tool offset on center of the VDI40 stations.

For Hybrid (BOT and VDI40 combination) turrets: Pressing **[F2]** sets an X-axis tool offset on center of the VDI40 stations.